Does a computer actually “know” how to compute

sin(x), e^x *or ln(x)?*

- A computer only knows:
  - Addition
  - Subtraction
  - Multiplication
  - Division
- → So how does it compute **complicated functions?**

Does a computer actually "know" how to compute **sin(x), e^x *or ln(x)?***

A) By accessing a massive database of pre-calculated values (lookup tables stored in its memory.)
B) By replacing them with polynomial functions that are easier to compute.
C) Computers have specialized hardware for every unique mathematical function.

Motivation

- The answer is:
  - We replace complicated functions with polynomials.
  - Why polynomials?
    - Because they are easy to compute.
  - And this is exactly what **Taylor's theorem** allows us to do.
  - It tells us that any sufficiently smooth function can be approximated near a point by a polynomial.

$$f(x) \approx \text{Polynomial}$$

Taylor Theorem

Chapter 4

Today's lecture is the mathematical backbone of Numerical Analysis.

Almost every algorithm we study later depends on this theorem.

Taylor Series

The Taylor series expansion of $f(x)$ about a:

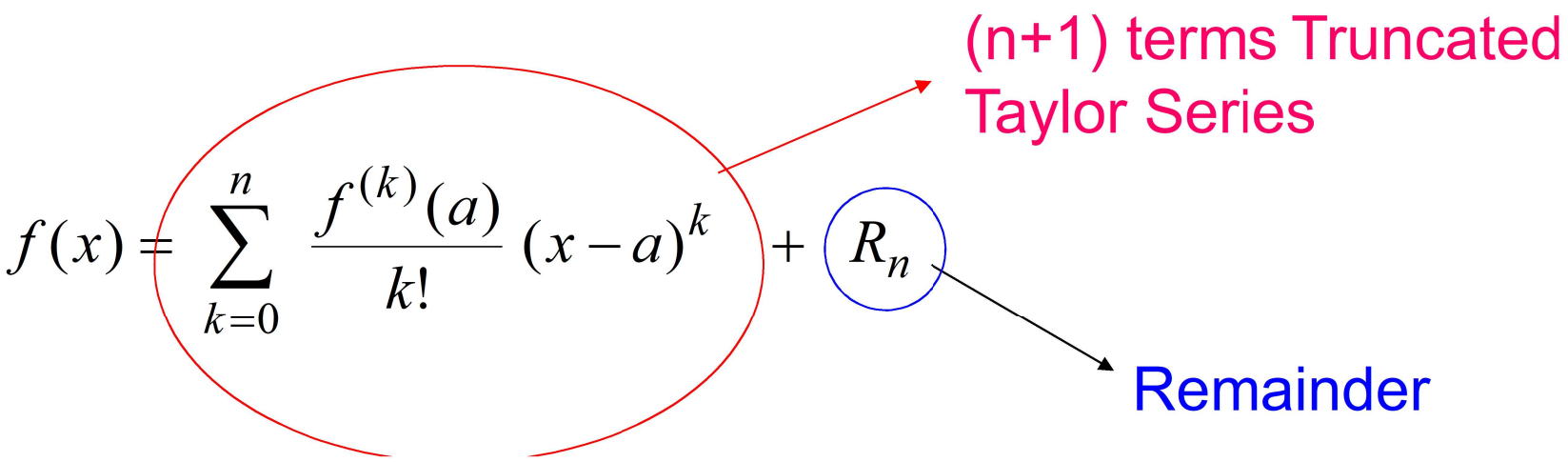
$$f(a)+f'(a)(x-a)+\frac{f^{(2)}(a)}{2!}(x-a)^2+\frac{f^{(3)}(a)}{3!}(x-a)^3+\ldots$$

or

$$f(x)=\sum_{k=0}^{\infty}\frac{1}{k!}f^{(k)}(a)\,(x-a)^k$$

Taylor's Theorem

If a function $f(x)$ possesses derivatives of orders $1, 2, ..., (n+1)$ on an interval containing a and x then the value of $f(x)$ is given by :



where the remainder R_n is defined as

$$R_n = \int_a^x \frac{(x-t)^n}{n!} f^{n+1}(t)dt$$

Taylor Theorem

Let $a = x_i$

$$f(x) \cong f(x_i) + \frac{f'(x_i)}{1!}(x - x_i) + \frac{f''(x_i)}{2!}(x - x_i)^2 + \ldots + \frac{f^{(n)}(x_i)}{n!}(x - x_i)^n + R_n$$

- ***zero order*** approximation:

$$f(x) \cong f(x_i)$$

- ***first order*** approximation:

$$f(x) \cong f(x_i) + f'(x_i)(x - x_i)$$

- ***Second order*** approximation:

$$f(x) \cong f(x_i) + \frac{f'(x_i)}{1!}(x - x_i) + \frac{f''(x_i)}{2!}(x - x_i)^2$$

- ***n^{th} order*** approximation

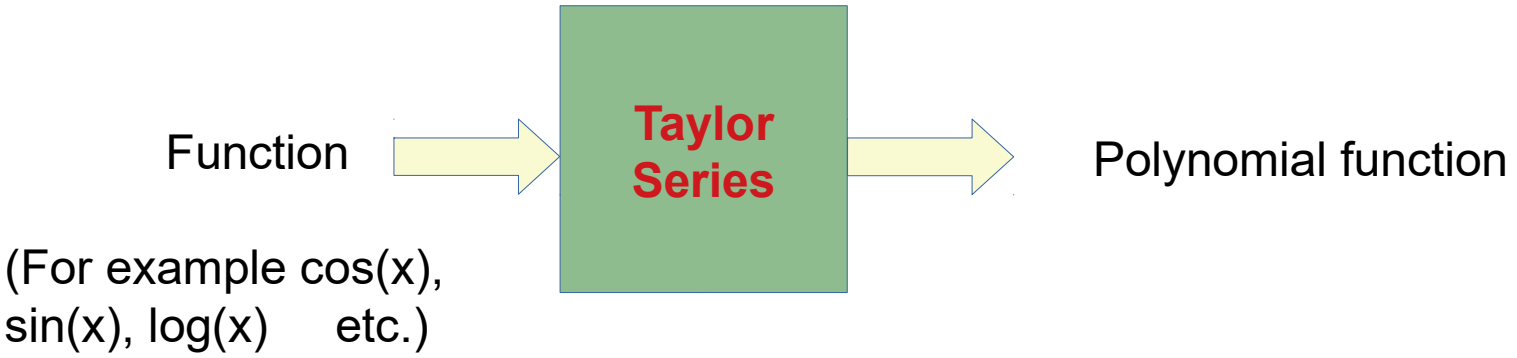
$$f(x) \cong f(x_i) + \frac{f'(x_i)}{1!}(x - x_i) + \frac{f''(x_i)}{2!}(x - x_i)^2 + \ldots + \frac{f^{(n)}(x_i)}{n!}(x - x_i)^n + R_n$$

Taylor's Theorem

- Taylor series can be used to predict the value of a function at one point in terms of the **function value and its derivatives at another point.**

Taylor's Theorem

- Taylor series allow us to approximate complicated functions with simpler polynomial functions.
- This simplification is particularly useful when dealing with functions that are difficult to integrate or differentiate directly.



Alternative form of Taylor's Theorem

Let $f(x)$ have derivatives of orders $1, 2, ..., (n+1)$ on an interval containing x and $x+h$ then :

$$f(x+h) = \sum_{k=0}^{n} \frac{f^{(k)}(x)}{k!} h^k + R_n \qquad (h = \text{step size})$$

$$R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1} \quad \textit{where } \xi \text{ is between } x \text{ and } x+h$$

Taylor's Theorem – Alternative forms

$$f(x)=\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k+\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$$

where ξ is between a and x.

$$a\to x,\quad x\to x+h$$

$$f(x+h)=\sum_{k=0}^{n}\frac{f^{(k)}(x)}{k!}h^k+\frac{f^{(n+1)}(\xi)}{(n+1)!}h^{n+1}$$

where ξ is between x and $x+h$.

Maclaurin Series

- Maclaurin series is a special case of Taylor series with the center of expansion ***a* = 0.**

The Maclaurin series expansion of $f(x)$:

$$f(0)+f'(0)x+\frac{f^{(2)}(0)}{2!}x^2+\frac{f^{(3)}(0)}{3!}x^3+\ldots$$

If the series converge, we can write:

$$f(x)=\sum_{k=0}^{\infty}\frac{1}{k!}f^{(k)}(0)\,x^k$$

Maclaurin Series – Example 1

Obtain Maclaurin series expansion of $f(x) = e^x$

$$f(0) + f'(0)x + \frac{f^{(2)}(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \ldots$$

Maclaurin Series – Example 1

Obtain Maclaurin series expansion of $f(x) = e^x$

$$f(x) = e^x \qquad f(0) = 1$$

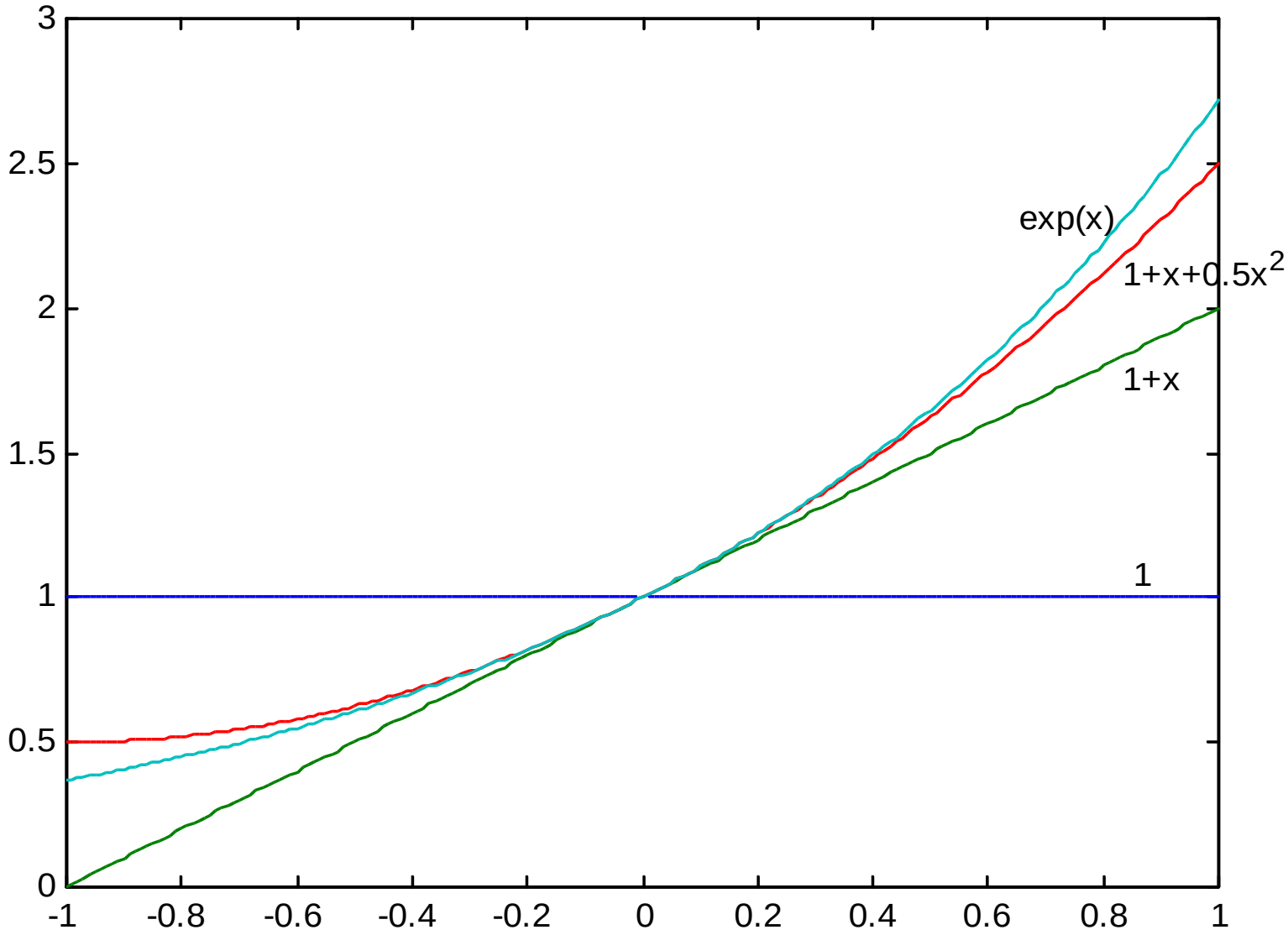
$$f'(x) = e^x \qquad f'(0) = 1$$

$$f^{(2)}(x) = e^x \qquad f^{(2)}(0) = 1$$

$$f^{(k)}(x) = e^x \qquad f^{(k)}(0) = 1 \quad for\, k \geq 1$$

$$e^x = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(0)\, x^k = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

The series converges for $|x| < \infty$.


3
2.5
2
1.5
1
0.5
0
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1
exp(x)
1+x+0.5x2
1+x
1

Example

Example: e^x for x=2

Using calculator **e^2 = 7.389056.**

But using Taylor series: $$f(x) \cong 1 + \frac{x}{1!} + \frac{x^2}{2!} + \ldots + \frac{x^n}{n!}$$

Using 4 terms: $$f(2) \cong 1 + \frac{2}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} = 6.333$$

Using 6 terms: $$f(2) \cong 1 + \frac{2}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!} + \frac{2^5}{5!} = 7.2666$$

➔ As more terms we use the closer we get.

Example

Using 4 terms: $$f(2) \cong 1 + \frac{2}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} = 6.333$$

Using 6 terms: $$f(2) \cong 1 + \frac{2}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!} + \frac{2^5}{5!} = 7.2666$$

using 5 terms → $f(2) = 6.999666$

Relative approximation error= ((7.2666- 6.999666)/7.2666) . 100 = 3.67 %

Using 6 terms if Using calculator $\mathbf{e^2 = 7.389056.}$

Relative true error= (7.389056-7.2666/ 7.389056) . 100 = 1.6 %

Remark

- In this course, all angles are assumed to be in **radian** unless you are told otherwise.

Maclaurin Series – Example 2

Obtain Maclaurin series expansion of $f(x) = \sin(x)$:

$$f(0) + f'(0)x + \frac{f^{(2)}(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \ldots$$

Maclaurin Series – Example 2

Obtain Maclaurin series expansion of $f(x) = \sin(x)$:

$$f(x) = \sin(x) \qquad f(0) = 0$$

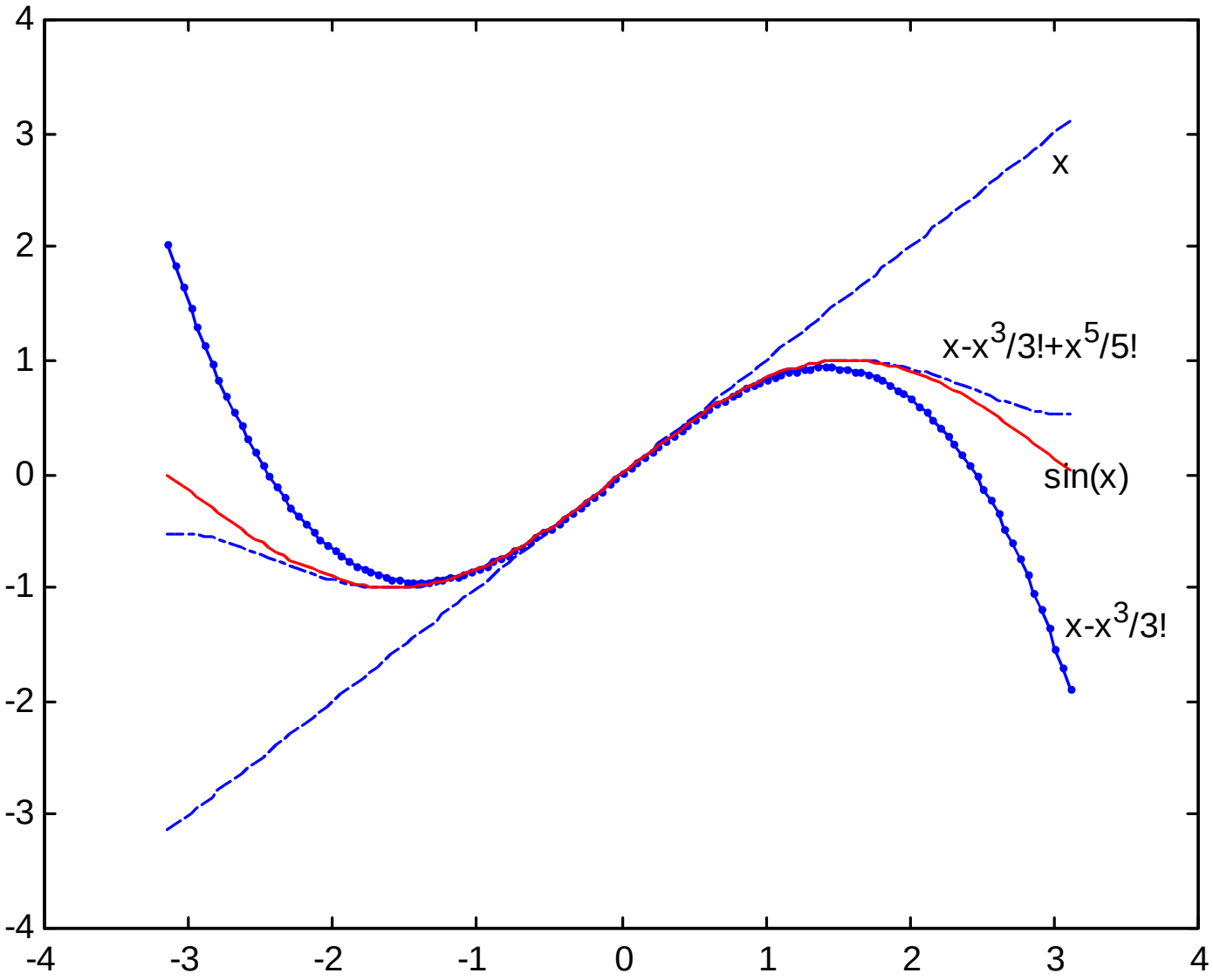
$$f'(x) = \cos(x) \qquad f'(0) = 1$$

$$f^{(2)}(x) = -\sin(x) \qquad f^{(2)}(0) = 0$$

$$f^{(3)}(x) = -\cos(x) \qquad f^{(3)}(0) = -1$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots.$$

The series converges for $|x| < \infty$.


4
3
2
1
0
-1
-2
-3
-4
-4
-3
-2
-1
0
1
2
3
4
x
x-x3/3!+x5/5!
sin(x)
x-x3/3!

Maclaurin Series – Example 3

Obtain Maclaurin series expansion of : $f(x) = \cos(x)$

$$f(0) + f'(0)x + \frac{f^{(2)}(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \ldots$$

Maclaurin Series – Example 3

Obtain Maclaurin series expansion of : $f(x)=\cos(x)$

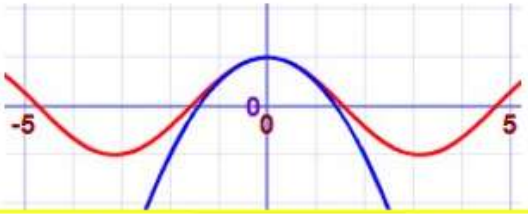
$$f(x)=\cos(x) \qquad f(0)=1$$
$$f'(x)=-\sin(x) \qquad f'(0)=0$$
$$f^{(2)}(x)=-\cos(x) \qquad f^{(2)}(0)=-1$$
$$f^{(3)}(x)=\sin(x) \qquad f^{(3)}(0)=0$$

$$\cos(x)=\sum_{k=0}^{\infty}\frac{f^{(k)}(0)}{k!}(x)^k=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+....$$

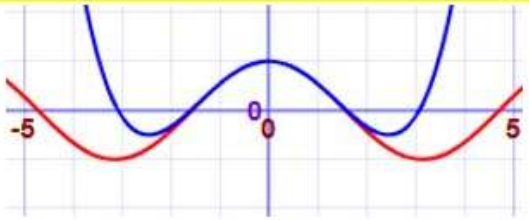
The series converges for $|x|<\infty$.

$$\cos(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}(x)^k = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots.$$

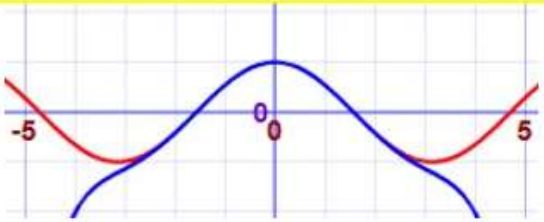
$$f(x) \cong 1 - \frac{x^2}{2!}$$



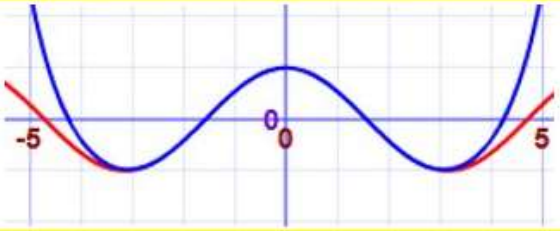
$$f(x) \cong 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$$



$$f(x) \cong 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}$$



$$f(x) \cong 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!}$$



Example

- Approximate the function ***f(x) = sin (x)*** from $x_0 = 0.2$ then find *f(0.6)* for a **relative true error** of less than **0.06%**. Use 4 significant digits in your calculation.
- Solution
  - a=0.2

$f(x) = \sin(x) \rightarrow f(0.2) = 0.1987$

$f'(x) = \cos(x) \rightarrow f'(0.2) = 0.9801$

$f''(x) = -\sin(x) \rightarrow f''(0.2) = -0.1987$

$f'''(x) = -\cos(x) \rightarrow f'''(0.2) = -0.9801$

Example

$f(x) = \sin(x) \rightarrow f(0.2) = 0.1987$

$f'(x) = \cos(x) \rightarrow f'(0.2) = 0.9801$

$f''(x) = -\sin(x) \rightarrow f''(0.2) = -0.1987$

$f'''(x) = -\cos(x) \rightarrow f'''(0.2) = -0.9801$

- *using **zero order** (1 term)* approximation:

$T_0 = f(x) \approx f(x_i) = f(0.2) = 0.1987$

- *using **first order** (2 terms)* approximation:

$$f(x) \cong f(x_i) + f'(x_i)(x - x_i)$$

$$T_1 = f(x) = 0.1987 + 0.9801(x - 0.2)$$

$$x = 0.6 \rightarrow T_1 = 0.5907$$

- using **second order** approximation

$$f(x) \cong f(x_i) + \frac{f'(x_i)}{1!}(x - x_i) + \frac{f''(x_i)}{2!}(x - x_i)^2$$

$$T_2 = f(x) = 0.198 + 0.9801(x - 0.2) - 0.1987(x - 0.2)^2$$

at x =0.6 → T_2 = 0.5748

- using **third order** approximation

$$f(x) \cong f(x_i) + \frac{f'(x_i)}{1!}(x - x_i) + \frac{f''(x_i)}{2!}(x - x_i)^2 + \frac{f'''(x_i)}{3!}(x - x_i)^3$$

$$T_3 = f(x) = 0.198 + 0.9801(x - 0.2) - 0.1987(x - 0.2)^2 - 0.9801(x - 0.2)^3$$

at x =0.6 → T_3 = 0.5643

True error $E_t = \text{true value} - \text{approximation}$

True percent relative error $\varepsilon_t = \dfrac{\text{true value} - \text{approximation}}{\text{true value}} 100\%$

The exact value sin(0.6) = 0.5646

(0.3659/0.5646) X 100= 29.6 %

Terms	$f(0.6)$	True error $E_t = \lvert f(0.6) - \sin(0.6) \rvert$	Relative true error
1	0.1987	0.3659	29.6%
2	*0.5907*	0.0261	4.62%
3	0.5748	0.0102	1.80 %
4	0.5643	0.0003	0.05 %

relative true error < 0.06% → stop

0.5646-0.1987=0.3659

Approximate percent relative error $$\varepsilon_a = \frac{\text{present approximation} - \text{previous approximation}}{\text{present approximation}} 100\%$$

To calculate the relative approximation error of the last example

Terms	F(0.6)	Relative approximation (estimated) Error
1	0.1987	-
2	0.5907	((0.5907-0.1987)/ 0.5907) X 100 = 66.36%
3	0.5748	**((0.5748-0.5907)/0.5748) X 100 = 2.76 %**
4	0.5643	**((0.5643-0.5748)/0.5643) X 100 = 1.86 %**

Example

Use zero- through ***fourth-order*** (*5 trails or terms)*Taylor series expansions to approximate the function estimate f(1.35) *with a= 0.75 (acceptance error 5%)*

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

- *solution*

Example

$a = 0.75$

$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$	$f(0.75)=1.2$
$f'(x)=$	
$f''(x)=$	
$f^3(x)=$	
$f^4(x) =$	

Example

$a = 0.75$

$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$	$f(0.75)=1.2$
$f'(x)=-0.4x^3-0.45x^2-x-0.25$	$f'(0.75)= ?$
$f''(x)= -1.2x^2-0.9x-1$	$f''(0.75) =?$
$f^3(x)= - 2.4 x -0.9$	$f^3(0.75) =?$
$f^4(x) =f^4(x)= -2.4$	$f^4(0.75) =?$

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

$$x=1.35,\ a=0.75 \rightarrow (x-a)=(1.35-0.75)=0.6$$

Trail	*f(x)*	*Approximate f(1.35)*	*App. Relative Error*
1	***f(x)=f(a)***		
2			
3			
4			
5			

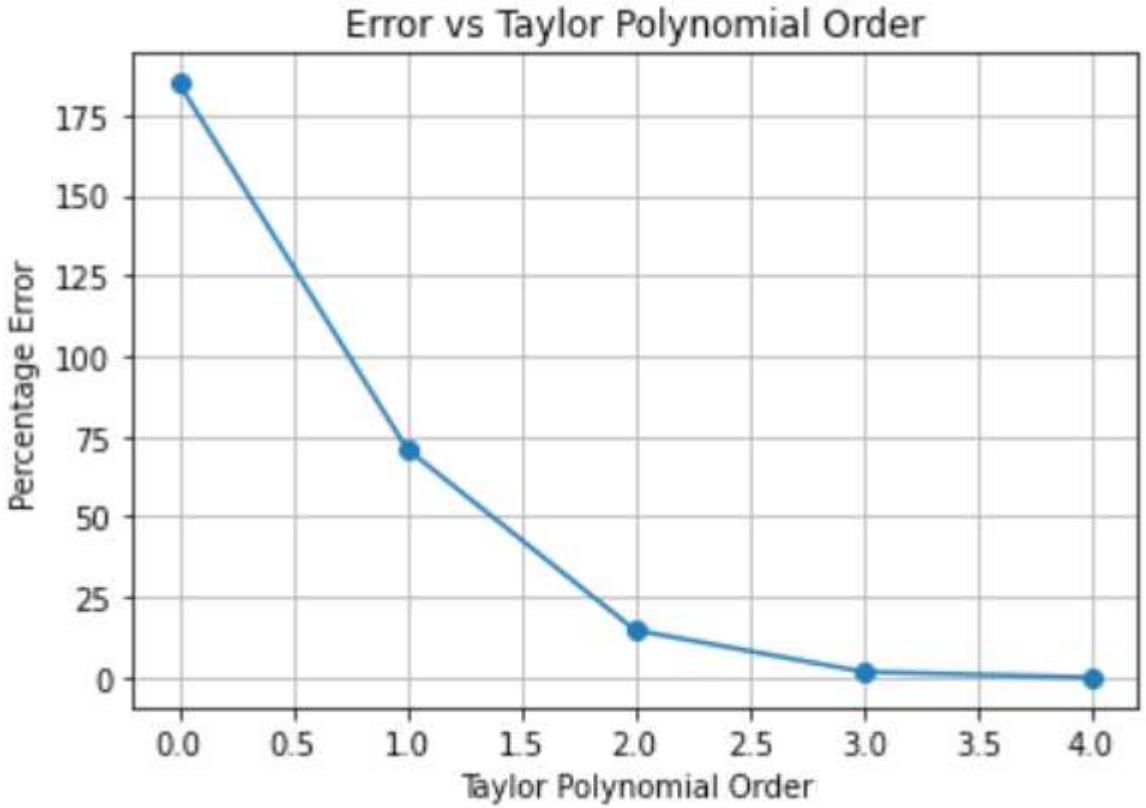
$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

$$x=1.35,\ a=0.75 \rightarrow (x-a)=(1.35-0.75)=0.6$$

	$f(x)$	Approximate $f(1.35)$	App. Relative Error
1	$f(x)=f(a)$	0.636328125 A	-
2	$f(x)=f(a)+f'(x).(x-a)$ $f(x)=A+f'(x).(x-a)$	-0.216796875 B	$\lvert B-A\rvert/B \cdot 100\% = 393.5\%$
3	$f(x)=B+f''(x).(x-a)^2/2!$	-0.639796857 C	$\lvert C-B\rvert/C \cdot 100\% = 66.11\ \%$
4	$f(x)=C+f'''(x).(x-a)^3/3!$	-0.736996875 D	$\lvert D-C\rvert/D \cdot 100\% = 13.1\ \%$
5	$f(x)=D+f''''(x).(x-a)^4/4!$	-0.749956875 E	$\lvert E-D\rvert/E \cdot 100\%$ $1.73\ \% < 5\%$

Exact value f(1.35) = -0.749956875000003

	f(x)	*Approximate f(1.35)*	***True. Relative Error***
1	$f(x)=f(a)$	*0.636328125* **A**	*184.85%*
2	$f(x)=f(a)+f'(x).(x-a)$ $f(x)=A+f'(x).(x-a)$	*-0.216796875* ***B***	**71.09%**
3	$f(x)=B+f''(x).(x-a)^2/2!$	*-0.639796857* ***C***	**14.69%**
4	$f(x)=C+f^3(x).(x-a)^3/3!$	*-0.736996875* ***D***	**1.73%**
5	$f(x)=D+f^4(x).(x-a)^4/4!$	*-0.749956875* ***E***	*0.00%*



Since the function is a 4th-degree polynomial,
the 4th-order Taylor expansion reproduces the function exactly.

Selected Problems from text book

- 4.1 – 4.2 – 4.3 – 4.4 – 4.5

Marking Scheme

- Midterm Exam: 40%
- Final Exam: 60%